

NOXUSDYNAMICS

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PROJECT REPORT

on

Automated Weed vs. Crop Detector Using Deep Learning

TEAM ID: NCD-5

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1. Abstract

The "Automated Weed vs. Crop Detector Using Deep Learning" addresses a critical challenge in modern precision agriculture: the automated, accurate discrimination of crops from invasive weeds. Traditional weed management relies heavily on manual inspection and uniform herbicide application, which is time-consuming, environmentally taxing, and economically inefficient. This project proposes an AI-powered image detection system utilizing the YOLO (You Only Look Once) real-time object detection algorithm. By processing field imagery, the system localizes plants via bounding boxes and classifies them as either "Crop" or "Weed" with corresponding confidence scores.

2. Problem Statement

In agricultural ecosystems, weeds compete directly with crops for essential resources including water, soil nutrients, sunlight, and physical growing space. The inability to rapidly and precisely identify weed locations leads to blanket herbicide spraying. This uniform application not only escalates operational costs but also contributes to soil degradation and chemical runoff. Therefore, a robust, automated computer-vision system is required to identify and localize both crops and weeds from agricultural field images quickly enough to facilitate targeted interventions.

3. Main Objectives

- **Automated Image Analysis:** Ingest and process standard agricultural field images.
- **Accurate Detection & Classification:** Detect all visible plants and accurately distinguish crop plants from weeds.
- **Precise Localization:** Generate bounding boxes around detected vegetation to indicate precise spatial locations.
- **Confidence Metrics:** Output statistical confidence scores for each classified bounding box to ensure operational reliability.
- **Practical Inference Speed:** Achieve processing speeds suitable for integration into real-world agricultural monitoring tools.

4. Proposed Solution & Methodology

The proposed system relies on a supervised deep learning approach, utilizing an annotated crop-and-weed image dataset to train a YOLO-based object detection model.

4.1 System Workflow

- **Image Input:** The system accepts an image of an agricultural field.
- **Image Preprocessing:** The image is resized, normalized, and formatted to match the input requirements of the YOLO network.
- **YOLO Object Detection:** The neural network processes the image in a single pass to identify distinct plant objects.
- **Classification:** Each localized object is evaluated and assigned to a defined class (e.g., 0 → Crop, 1 → Weed).
- **Localization & Output:** The system draws bounding boxes around the classified objects, overlaying the predicted label and the confidence percentage.

4.2 Algorithm Selection: YOLO

YOLO (You Only Look Once) was selected over traditional two-stage detectors (like Faster R-CNN) due to its optimized architecture that frames object detection as a single regression problem. This allows the model to process images in real-time while simultaneously predicting multiple bounding boxes and class probabilities, making it highly effective for dynamic agricultural environments.

5. Dataset and Preprocessing

The foundation of the model's accuracy is an annotated image dataset containing diverse examples of agricultural fields, specific crop types, and various weed species.

- **Annotation Format:** Bounding-box annotations are formatted specifically for YOLO training, utilizing the structure: `class_id x_center y_center width height` (with spatial coordinates normalized between 0 and 1).

Dataset Selection and Split

The project uses a single annotated crop-and-weed dataset containing agricultural field images with bounding-box annotations for two classes: Crop (Class 0) and Weed (Class 1). The dataset provided the required image and annotation format for training the YOLO-based object detection model.

Instead of combining multiple datasets, the project continued with the same dataset and divided it into training, validation, and testing subsets. This maintained consistency in the image characteristics, annotation format, class definitions, and data distribution throughout the training and evaluation process.

The dataset was divided approximately as follows:

- **70% Training:** Used to train the YOLO model to learn crop and weed features.
- **20% Validation:** Used to monitor model performance and tune the model during development.
- **10% Testing:** Used for the final evaluation on unseen images.

The 70:20:10 split provided a sufficiently large portion of the dataset for training while maintaining separate data for validation and final testing. In future, additional datasets containing different crop varieties, weed species, lighting conditions, and field environments can be incorporated to improve the model's ability to generalize to diverse real-world conditions.

6. Technology Stack

- **Programming Language:** Python
- **Deep Learning Architecture:** YOLO (Ultralytics implementation)
- **Machine Learning Framework:** PyTorch
- **Computer Vision Libraries:** OpenCV, NumPy, Matplotlib
- **Development & Training Environment:** Google Colab (utilizing GPU acceleration for efficient model training)

7. Expected Results & Evaluation Metrics

The final trained model is expected to process raw field imagery and output annotated images clearly distinguishing crops from weeds. The system's performance will be quantified using standard object-detection metrics:

- **Precision:** The ratio of true positive detections to the total number of positive predictions.
- **Recall:** The proportion of actual plants correctly identified by the model.
- **Mean Average Precision (mAP):** The comprehensive metric evaluating the accuracy of the bounding boxes across all classes.
- **Intersection over Union (IoU):** The spatial overlap between the predicted bounding box and the ground-truth annotation.

Model Evaluation Results

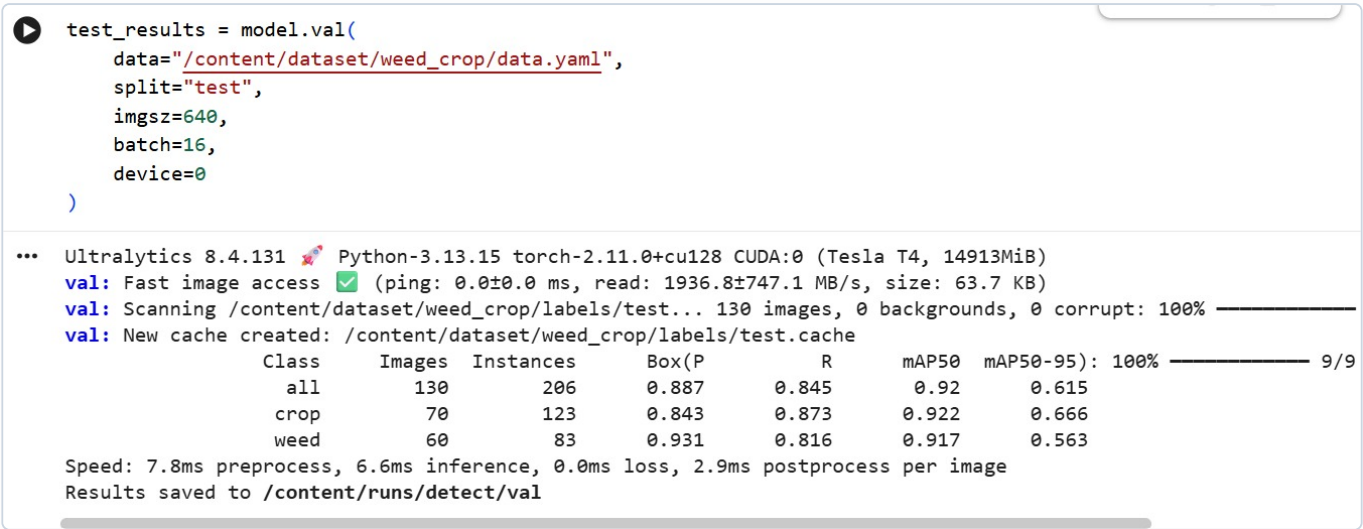


Figure 1: Model Evaluation Results Output

8. Project Development Timeline and Progress

4-Week Project Progress

Week 1 — Dataset Preparation

The crop and weed dataset was selected and its structure was analyzed. The dataset contained two classes: Class 0 – Crop and Class 1 – Weed. The annotations were verified to follow the YOLO format: class_id x_center y_center width height. The dataset was divided into 70% training, 20% validation, and 10% testing sets. The prepared dataset was then organized for YOLO model training.

Week 2 — Model Training

The model was trained using Google Colab with Ultralytics. GPU acceleration using a Tesla T4 was utilized to improve the training process. The YOLO11 model was selected and trained using the prepared crop and weed dataset. The trained model and its results were saved for further evaluation.

Week 3 — Testing and Evaluation

The trained model was tested using unseen test images to evaluate its performance. The evaluation included Precision, Recall, mAP@50, mAP@50–95, confusion matrix, and bounding-box detection.

The obtained evaluation results were:

- **Precision:** 88.7%
- **Recall:** 84.5%
- **mAP@50:** 92.0%
- **mAP@50–95:** 61.5%

The test set consisted of 130 images containing 206 annotated instances.

Week 4 — Documentation and Future Development

The methodology, dataset preparation, training process, and evaluation results were documented. The limitations of the current model were also identified, including small weeds, variations in lighting, occlusion, limited dataset diversity, and unseen plant species.

Future development will focus on real-world deployment using a camera, Edge-AI device, drone or rover, with possible applications in smart precision spraying and IoT/robotics-based agricultural systems.

9. Limitations

While highly capable, the current system is constrained by standard computer vision limitations. Detection accuracy is heavily dependent on the diversity and quality of the training dataset. Extreme variations in environmental lighting, severe plant occlusion, or the presence of microscopic weeds may degrade performance. Furthermore, deploying the model in fields with plant species entirely unrepresented in the training data will result in reduced classification accuracy.

10. Future Scope

The successful deployment of this automated detector lays the groundwork for expansive precision agriculture applications:

- **Drone Integration:** Mounting the camera and inference hardware on UAVs for rapid, large-scale field mapping.
- **Precision Spraying Hardware:** Coupling the detection software with automated smart-sprayers to deploy herbicide exclusively on identified weed coordinates.
- **IoT & Robotics:** Integrating the model into autonomous terrestrial agricultural rovers for continuous field monitoring and mechanical weed removal.